

Canonical GHZ configurations - cluster P3 - helicity conserving

species: electron
polarization cluster: P3
source cluster configurations: 87
canonical configurations collected: 10

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity conserving
classification: complete-chain R_flip/conserves ≤ 1
per-page labels: isolated flip/conserves and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 53 configurations
- 4 components: 8 configurations
- 5 components: 4 configurations
- 6 components: 19 configurations
- 7 components: 3 configurations

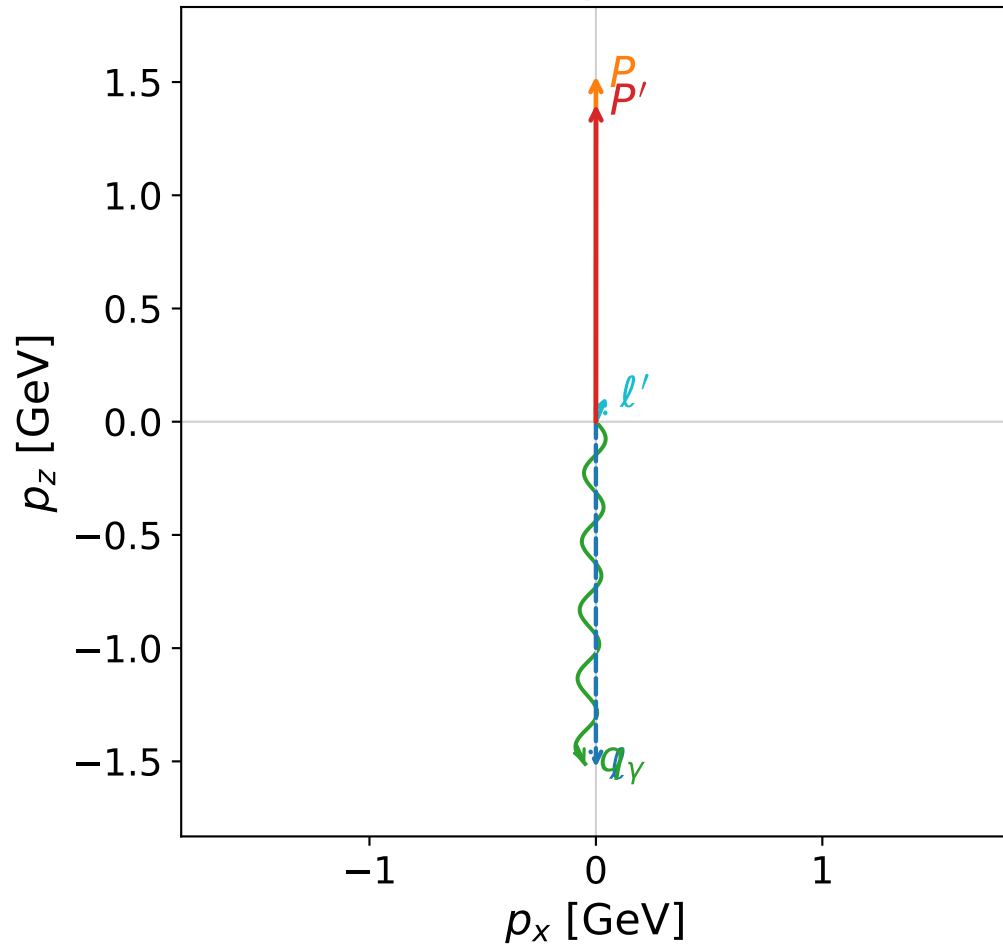
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_03\combined\
helicity_conserving\canonical_dghz_polarization_cluster_03_helicity_conserving_configu
rations_electron.csv

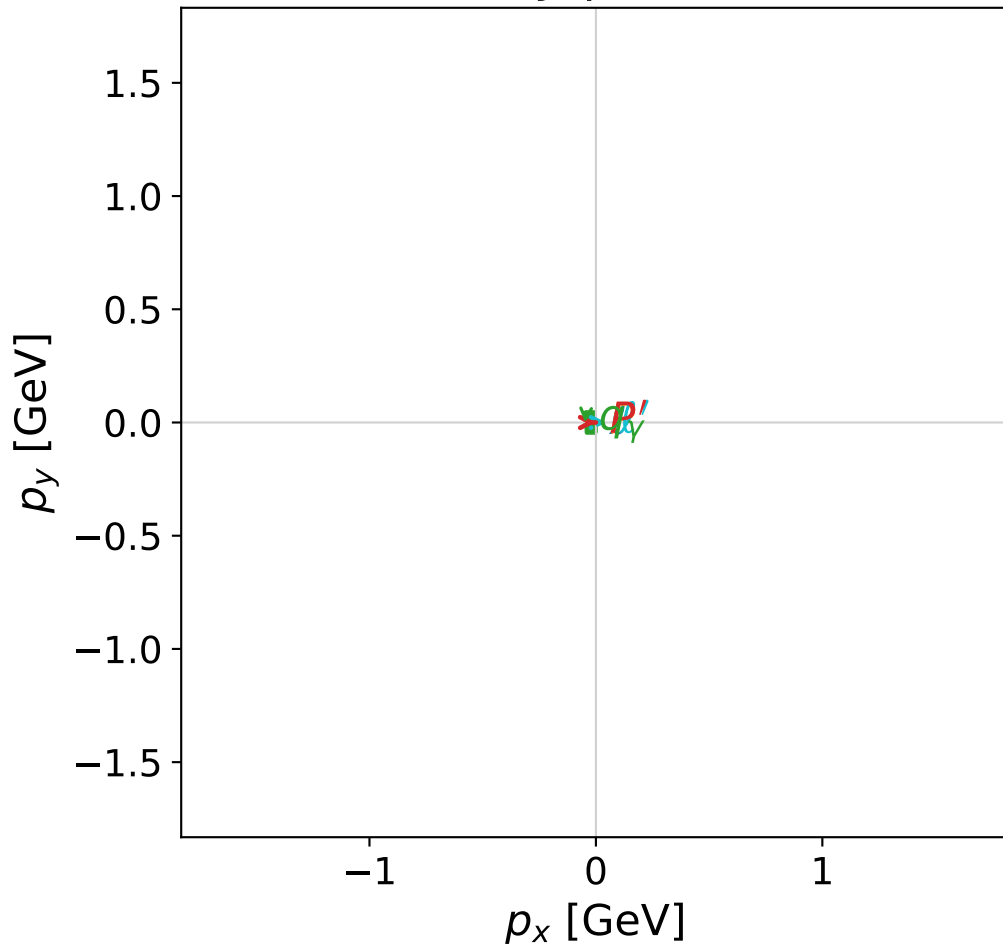
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

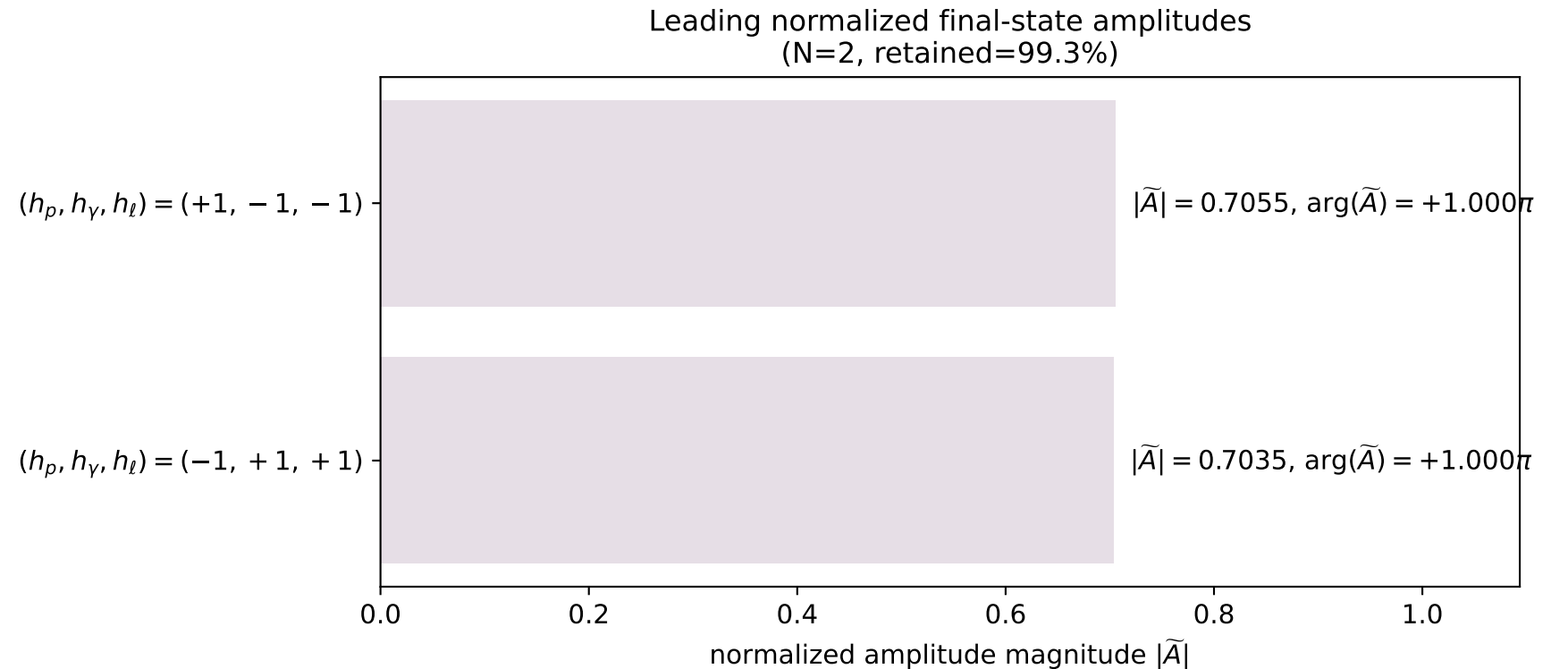


dghz_mixing_angles_polarization_03_local_minimum_188
 $d_{\{\text{rm GHZ}\}}=1.11719\text{e-}05$
 region: polarization_cluster_03_local_minimum_188
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0188
 lepton mixing angle: $\alpha_e=2.3764974$ rad
 incoming lepton: $-0.721316|+\rangle +0.692606|-\rangle$
 proton mixing angle: $\alpha_p=0.84717575$ rad
 incoming proton: $0.662102|+\rangle +0.749413|-\rangle$

$s=11.0066$, $\sqrt{s}=3.31762$, $|\vec{P}|=1.52621$, $|\vec{P}'|=1.40088$
 $\theta_{p'}=0.000417261$, $\theta_\gamma=3.11038$, $\phi_{p'}=6.35544\text{e-}05$, $\phi_\gamma=3.14165$
 $E_\gamma=1.51194$, $\theta_{\ell'}=0.399604$, $\phi_{\ell'}=6.18232\text{e-}05$
 $Q^2=0.702332$, $x_B=0.0699923$, $t=-0.00457805$, $W^2=10.2119$, $y=0.990883$

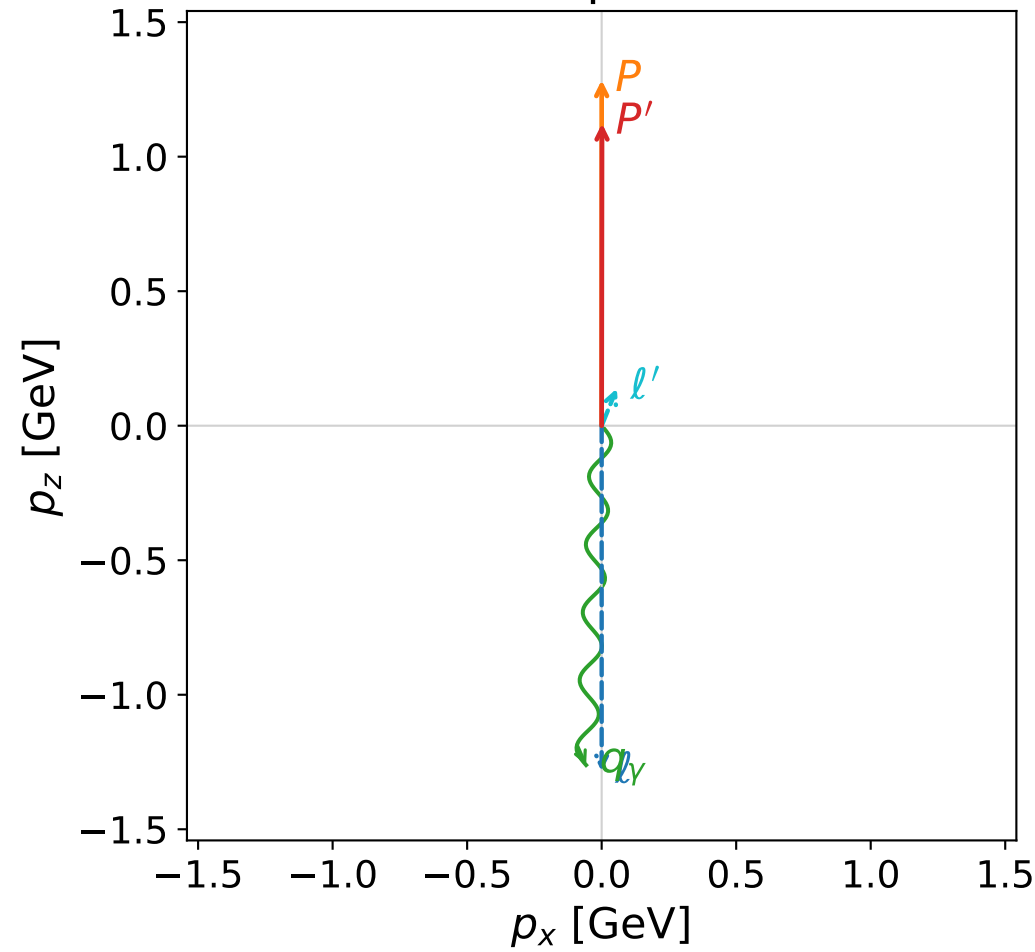
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5262$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5262$
 P $E= 1.7914$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5262$
 ℓ' $E= 0.1198$ $p_x= 0.0466$ $p_y= 0.0000$ $p_z= 0.1103$
 P' $E= 1.6859$ $p_x= 0.0006$ $p_y= 0.0000$ $p_z= 1.4009$
 q_γ $E= 1.5119$ $p_x= -0.0472$ $p_y= -0.0000$ $p_z= -1.5112$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.012%, conserve 99.988%
 $R_{\text{flip/conserve}}=1.164\text{e-}04$; $I_h=-1.571\text{e-}05$
 virtual-photon powers: T 99.590%, L 0.410%
 $R_L/T=4.119\text{e-}03$; $I_{TL}=+1.472\text{e-}03$

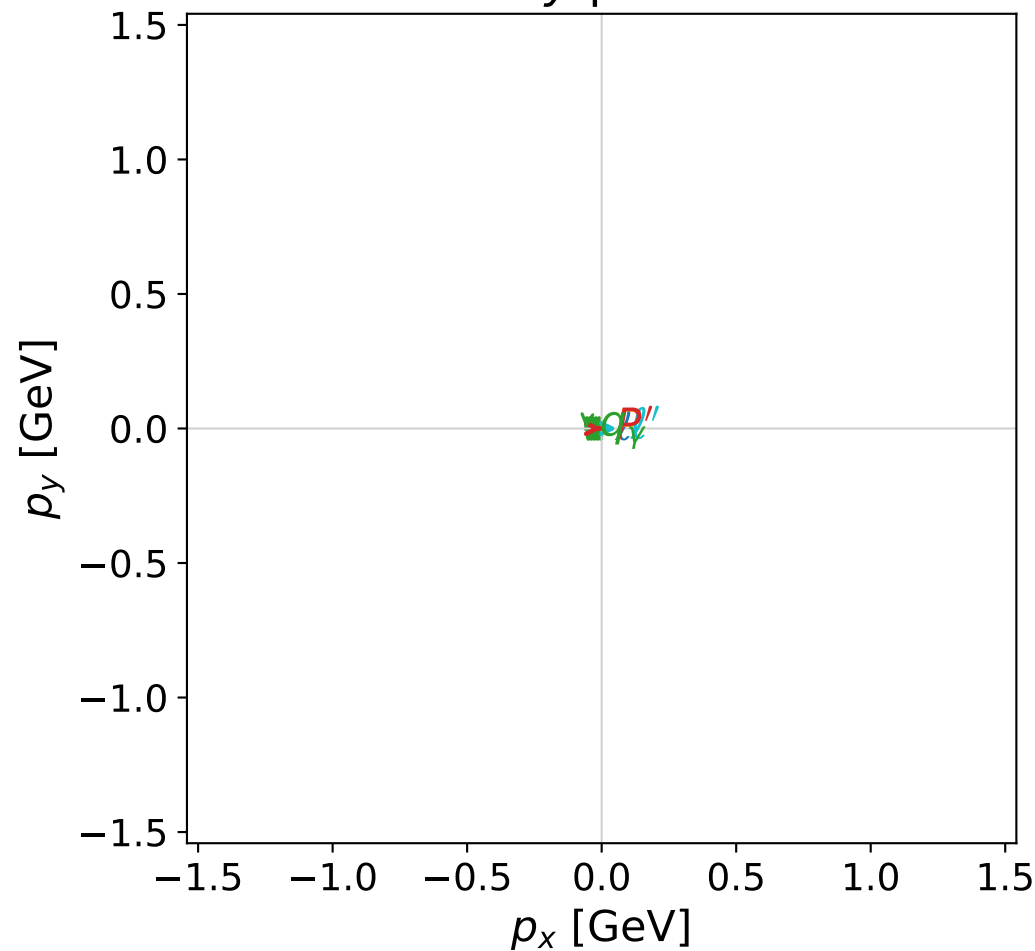


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

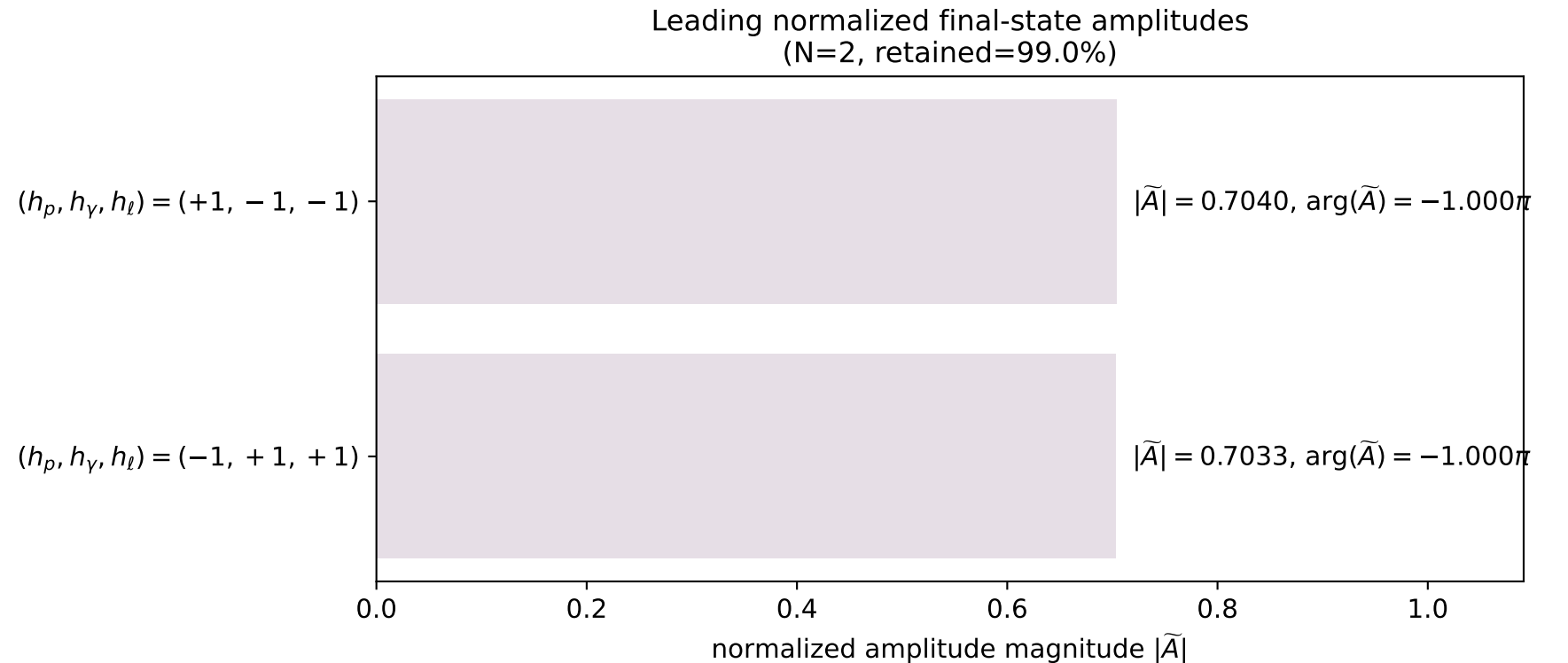


dghz_mixing_angles_polarization_03_local_minimum_189
 $d_{\text{GHZ}}=1.25241\text{e-}05$
 region: polarization_cluster_03_local_minimum_189
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0189
 lepton mixing angle: $\alpha_e=2.4349155$ rad
 incoming lepton: $-0.760524|+\rangle +0.64931|-\rangle$
 proton mixing angle: $\alpha_p=0.90852471$ rad
 incoming proton: $0.61491|+\rangle +0.788597|-\rangle$

$s=8.26616$, $\sqrt{s}=2.87509$, $|\vec{P}|=1.28454$, $|\vec{P}'|=1.12322$
 $\theta_{p'}=0.00126895$, $\theta_\gamma=3.09582$, $\phi_{p'}=6.28313$, $\phi_\gamma=3.14157$
 $E_\gamma=1.2626$, $\theta_{l'}=0.38751$, $\phi_{l'}=6.28316$
 $Q^2=0.737755$, $x_B=0.101526$, $t=-0.00984887$, $W^2=7.40874$, $y=0.983798$

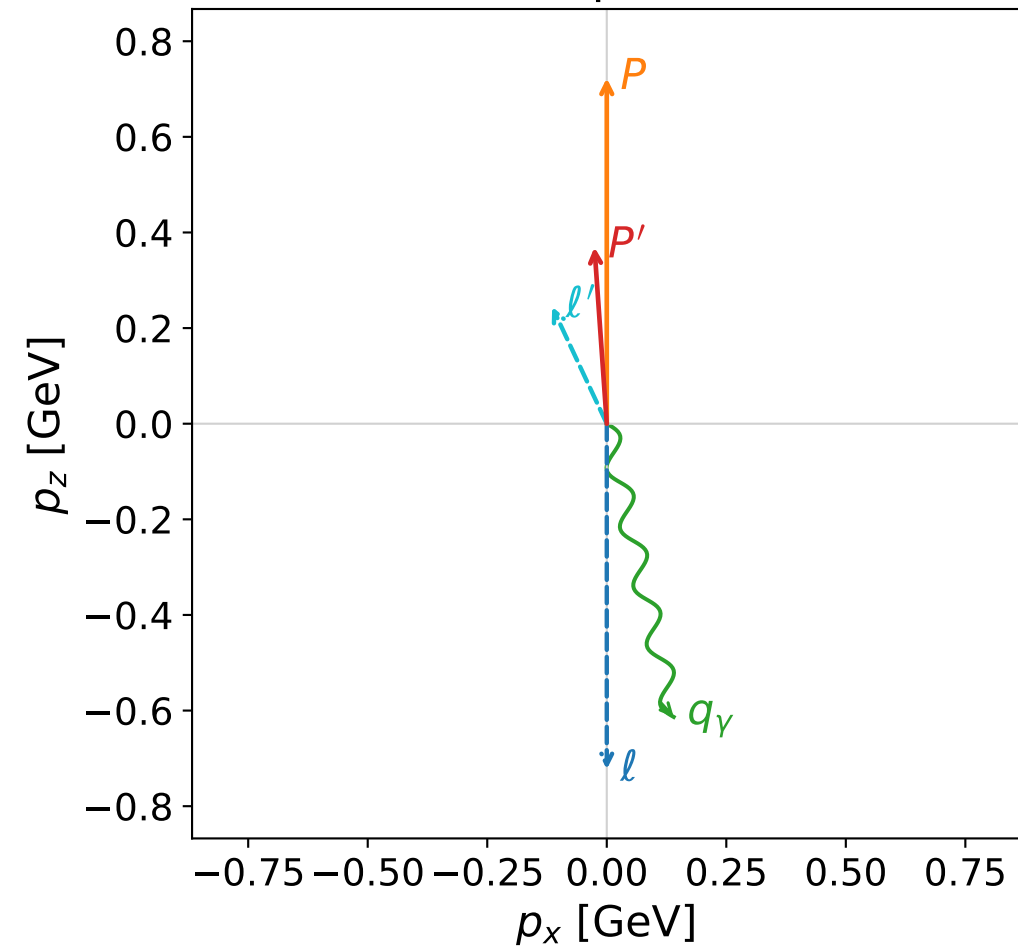
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.2845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2845$
 P $E= 1.5906$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2845$
 l' $E= 0.1491$ $p_x= 0.0563$ $p_y= -0.0000$ $p_z= 0.1381$
 P' $E= 1.4634$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1232$
 q_γ $E= 1.2626$ $p_x= -0.0578$ $p_y= 0.0000$ $p_z= -1.2613$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.008%, conserve 99.992%
 $R_{\text{flip/conserve}}=7.768\text{e-}05$; $I_h=-1.596\text{e-}05$
 virtual-photon powers: T 99.425%, L 0.575%
 $R_L/T=5.780\text{e-}03$; $I_{TL}=+4.576\text{e-}03$

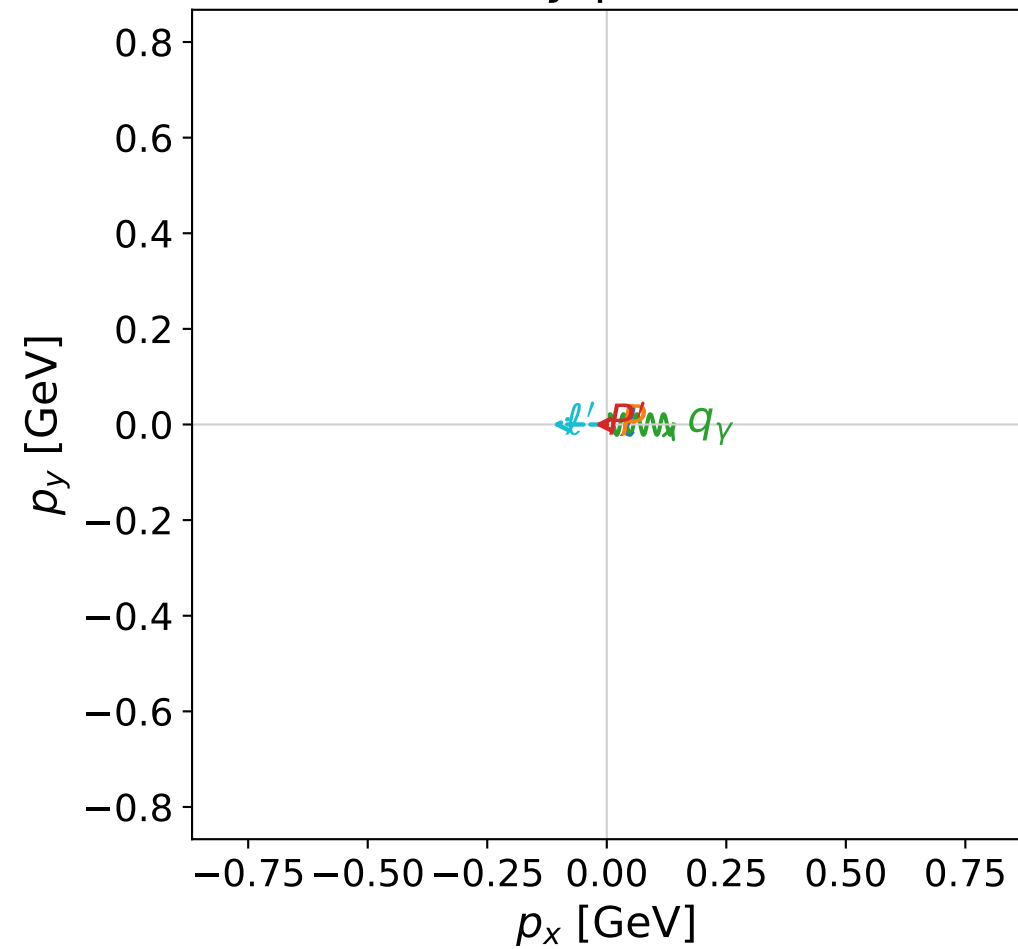


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_204

$d_{\{\text{rm GHZ}\}}=2.47799\text{e-}05$

region: polarization_cluster_03_local_minimum_204

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0204

lepton mixing angle: $\alpha_e=2.0175873$ rad

incoming lepton: $-0.432074|+\rangle +0.901838|-\rangle$

proton mixing angle: $\alpha_p=0.38602922$ rad

incoming proton: $0.926411|+\rangle +0.376513|-\rangle$

$s=3.63688$, $\sqrt{s}=1.90706$, $|\vec{P}|=0.72285$, $|\vec{P}'|=0.370003$

$\theta_{p'}=0.0696716$, $\theta_\gamma=2.91723$, $\phi_{p'}=3.14156$, $\phi_\gamma=0$

$E_\gamma=0.629105$, $\theta_{\ell'}=0.437411$, $\phi_{\ell'}=3.1416$

$Q^2=0.742871$, $x_B=0.300568$, $t=-0.0948674$, $W^2=2.60853$, $y=0.896454$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.7228$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7228$

P $E= 1.1842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7228$

ℓ' $E= 0.2696$ $p_x= -0.1142$ $p_y= -0.0000$ $p_z= 0.2442$

P' $E= 1.0083$ $p_x= -0.0258$ $p_y= 0.0000$ $p_z= 0.3691$

q_γ $E= 0.6291$ $p_x= 0.1400$ $p_y= 0.0000$ $p_z= -0.6133$

GHZ mechanism: helicity conserving (full H G C)

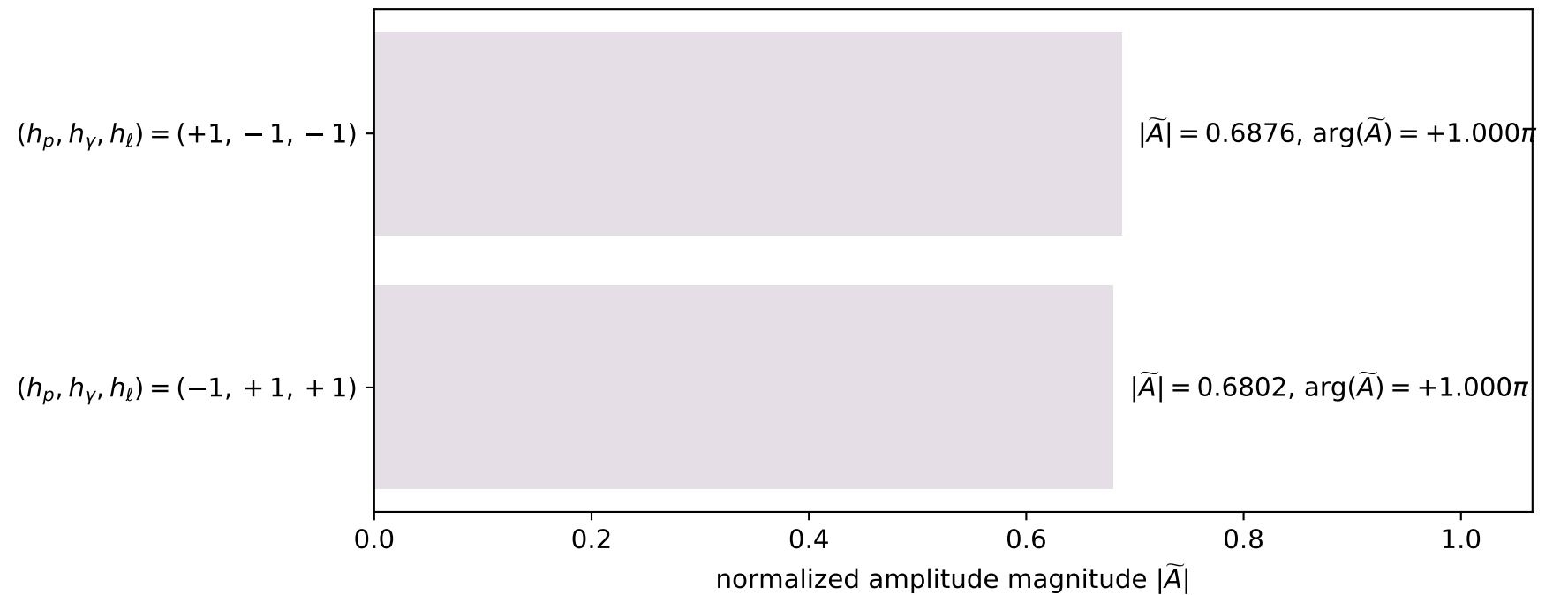
helicity powers: flip 0.002%, conserve 99.998%

$R_{\text{flip/conserve}}=1.743\text{e-}05$; $I_h=+6.405\text{e-}05$

virtual-photon powers: T 94.991%, L 5.009%

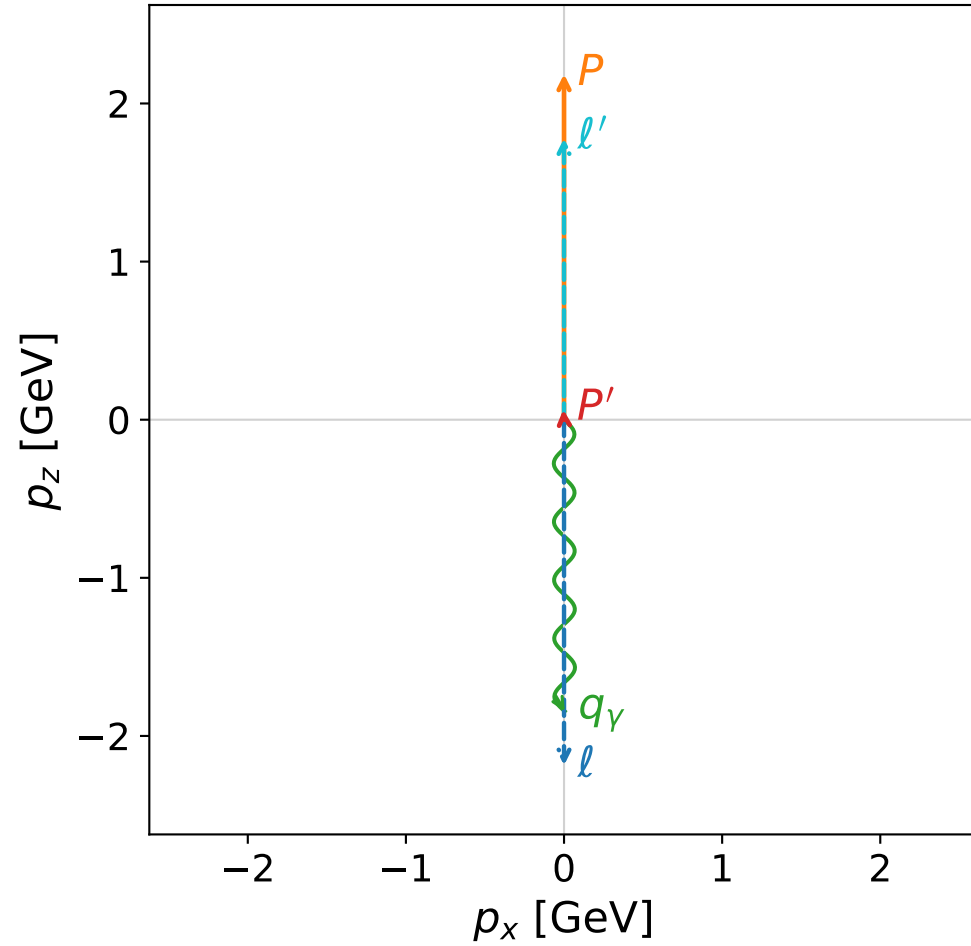
$R_L/T=5.273\text{e-}02$; $I_{TL}=+5.792\text{e-}02$

Leading normalized final-state amplitudes
(N=2, retained=93.6%)

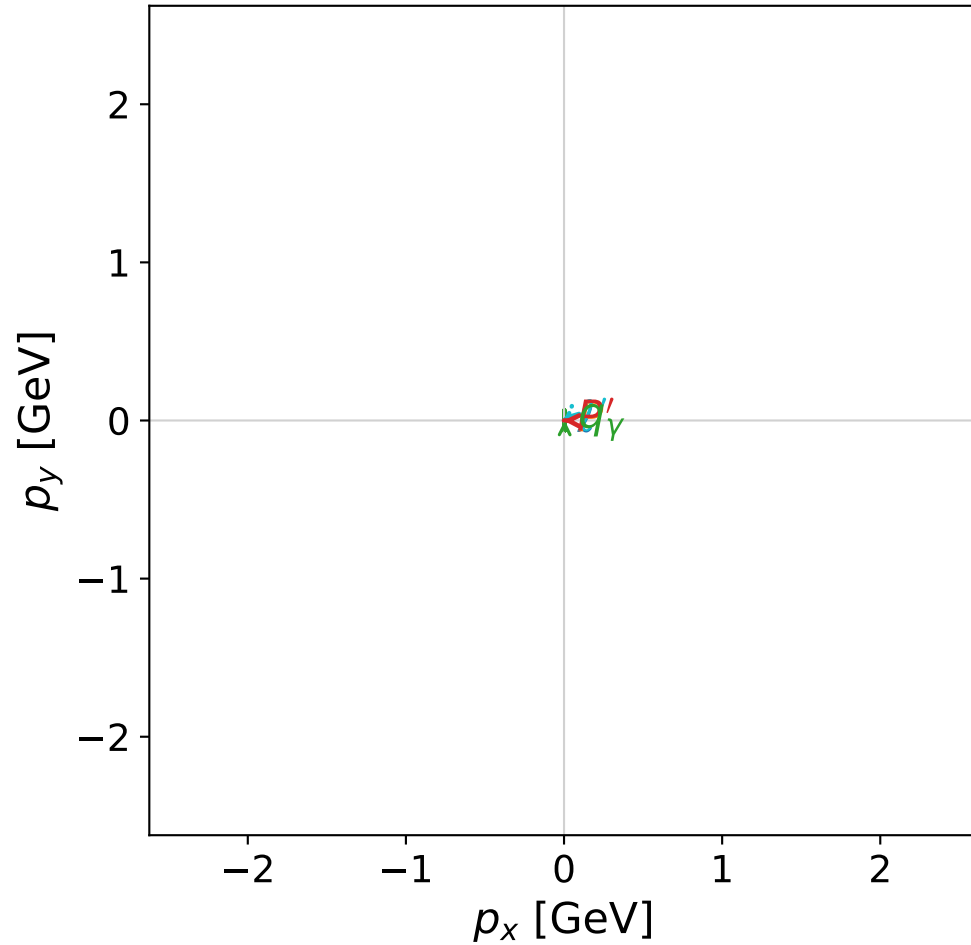


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



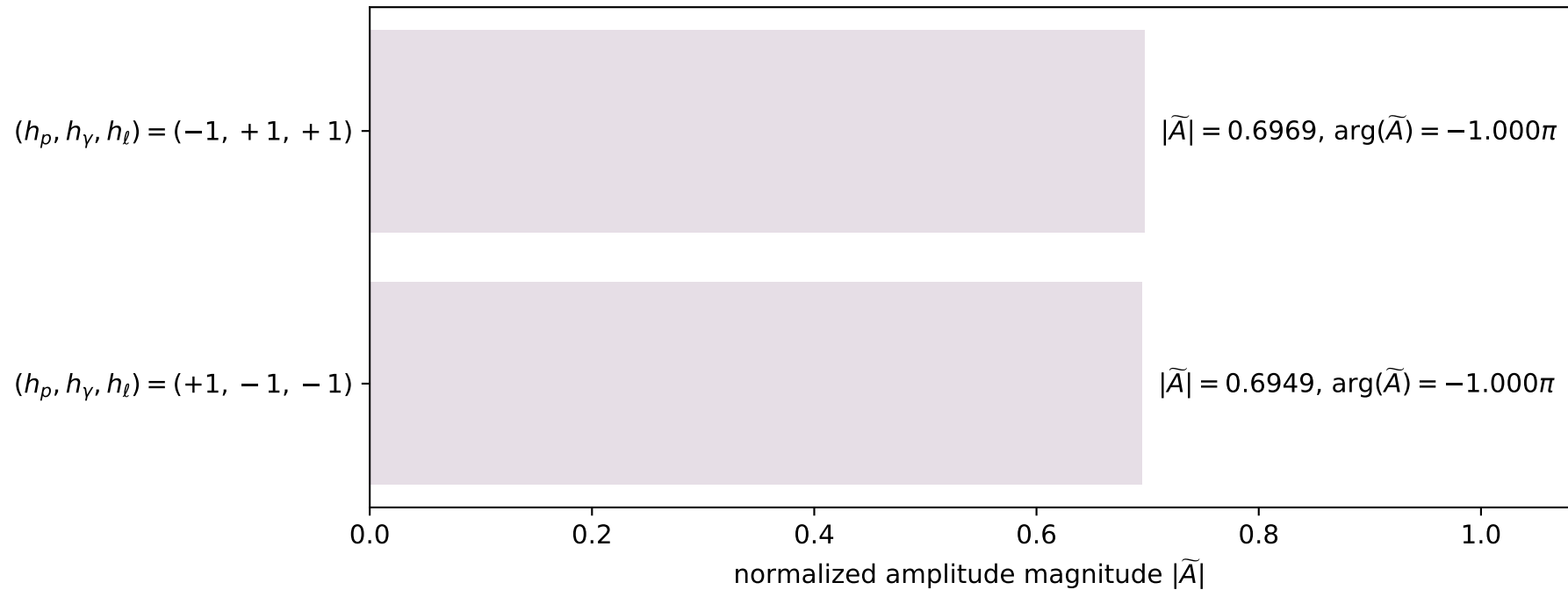
dghz_mixing_angles_polarization_03_local_minimum_248
 $d_{\text{GHZ}}=0.000127763$
 region: polarization_cluster_03_local_minimum_248
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0248
 lepton mixing angle: $\alpha_e=2.9049644$ rad
 incoming lepton: $-0.972134|+\rangle +0.234426|-\rangle$
 proton mixing angle: $\alpha_p=1.3339106$ rad
 incoming proton: $0.234676|+\rangle +0.972074|-\rangle$

$s=20.8306$, $\sqrt{s}=4.56405$, $|\vec{P}|=2.18564$, $|\vec{P}'|=0.064417$
 $\theta_{P'}=0.0689908$, $\theta_\gamma=3.13913$, $\phi_{P'}=3.12116$, $\phi_\gamma=0$
 $E_\gamma=1.84406$, $\theta_{\ell'}=7.53344\text{e-}05$, $\phi_{\ell'}=3.88476$
 $Q^2=15.5599$, $x_B=0.807696$, $t=-2.43181$, $W^2=4.58449$, $y=0.965605$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1856$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1856$
 P $E= 2.3784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1856$
 ℓ' $E= 1.7798$ $p_x= -0.0001$ $p_y= -0.0001$ $p_z= 1.7798$
 P' $E= 0.9402$ $p_x= -0.0044$ $p_y= 0.0001$ $p_z= 0.0643$
 q_γ $E= 1.8441$ $p_x= 0.0045$ $p_y= 0.0000$ $p_z= -1.8441$

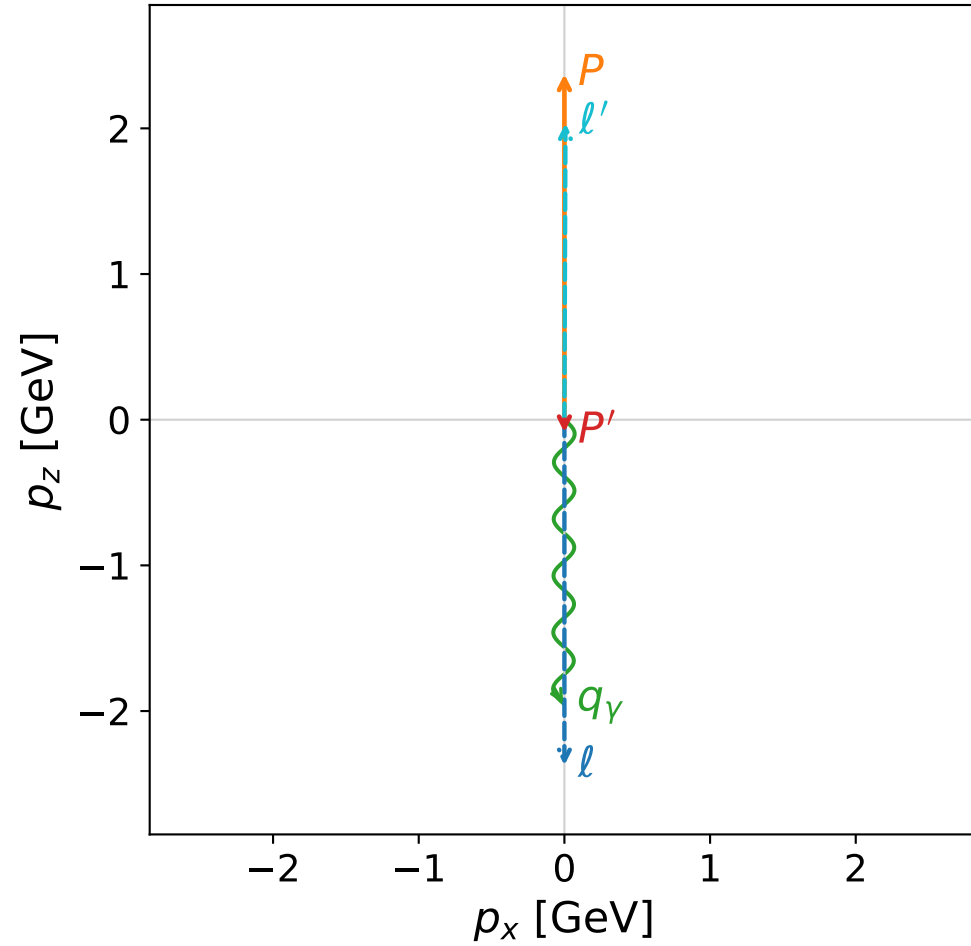
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 5.119%, conserve 94.881%
 $R_{\text{flip/conserve}}=5.395\text{e-}02$; $I_h=-4.518\text{e-}02$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.138\text{e-}07$; $I_{TL}=-4.601\text{e-}04$

Leading normalized final-state amplitudes
 (N=2, retained=96.9%)

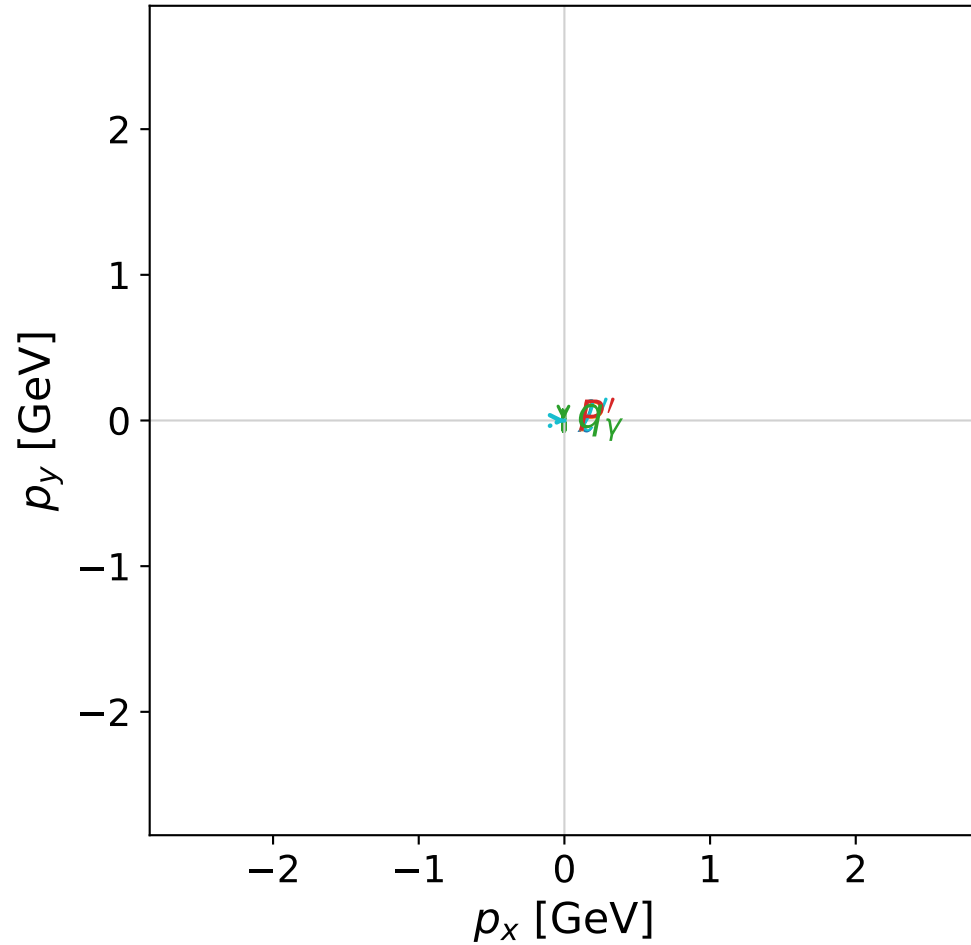


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

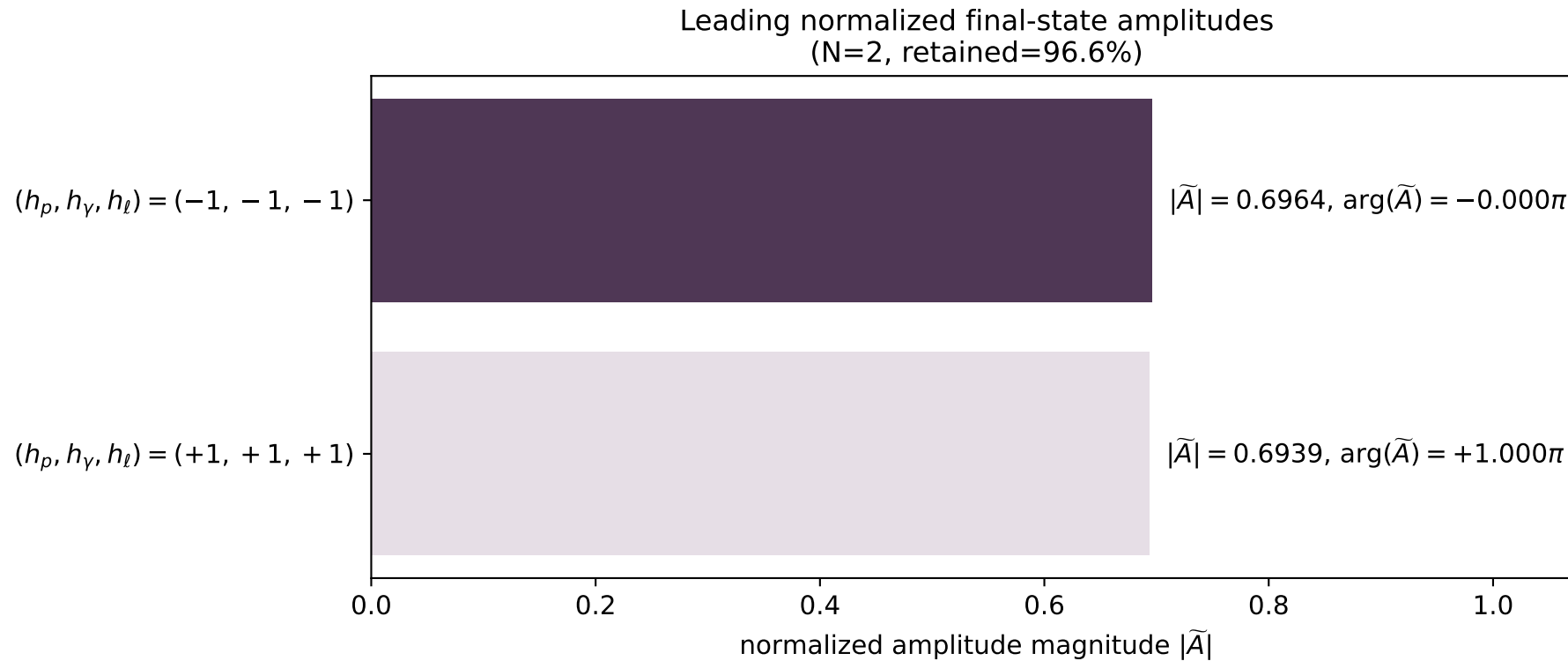


dghz_mixing_angles_polarization_03_local_minimum_262
 $d_{\text{GHZ}}=0.000177709$
 region: polarization_cluster_03_local_minimum_262
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0262
 lepton mixing angle: $\alpha_e=1.6990512$ rad
 incoming lepton: $-0.127904|+\rangle +0.991787|-\rangle$
 proton mixing angle: $\alpha_p=0.12870773$ rad
 incoming proton: $0.991729|+\rangle +0.128353|-\rangle$

$s=24.2413$, $\sqrt{s}=4.92355$, $|\vec{P}|=2.37242$, $|\vec{P}'|=0.0876034$
 $\theta_{P'}=3.14159$, $\theta_Y=3.13785$, $\phi_{P'}=3.00433$, $\phi_Y=3.14156$
 $E_Y=1.94693$, $\theta_{\ell'}=0.00358498$, $\phi_{\ell'}=6.28316$
 $Q^2=19.307$, $x_B=0.853001$, $t=-3.46271$, $W^2=4.20707$, $y=0.968871$

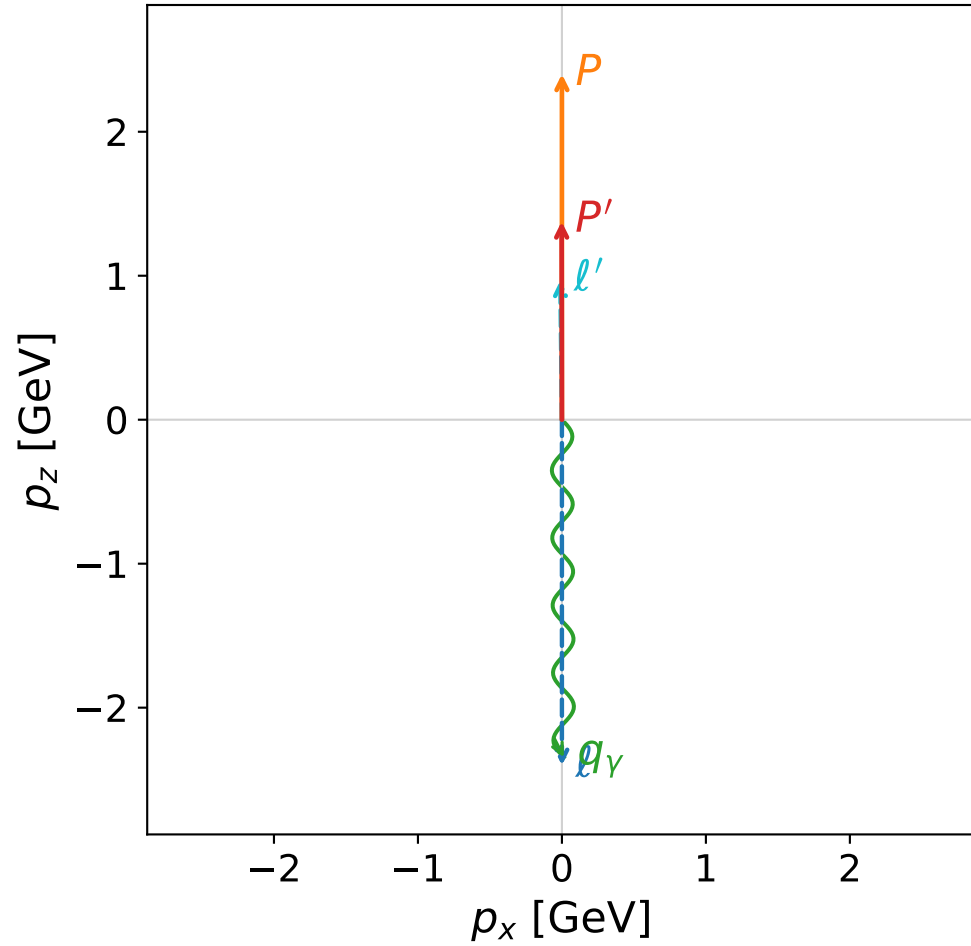
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3724$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3724$
 P $E= 2.5511$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3724$
 ℓ' $E= 2.0345$ $p_x= 0.0073$ $p_y= -0.0000$ $p_z= 2.0345$
 P' $E= 0.9421$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0876$
 q_Y $E= 1.9469$ $p_x= -0.0073$ $p_y= 0.0000$ $p_z= -1.9469$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 6.070%, conserve 93.930%
 $R_{\text{flip/conserve}}=6.462\text{e-}02$; $I_h=-5.850\text{e-}02$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.649\text{e-}07$; $I_{TL}=+1.885\text{e-}04$

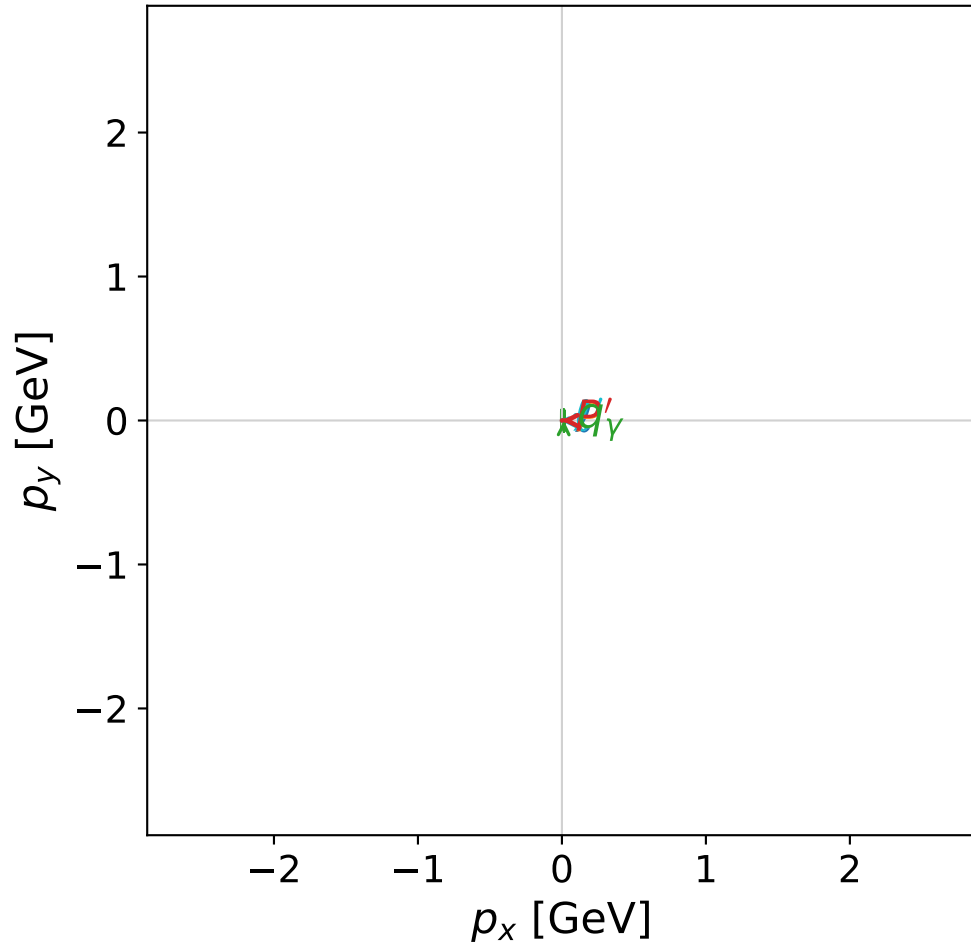


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

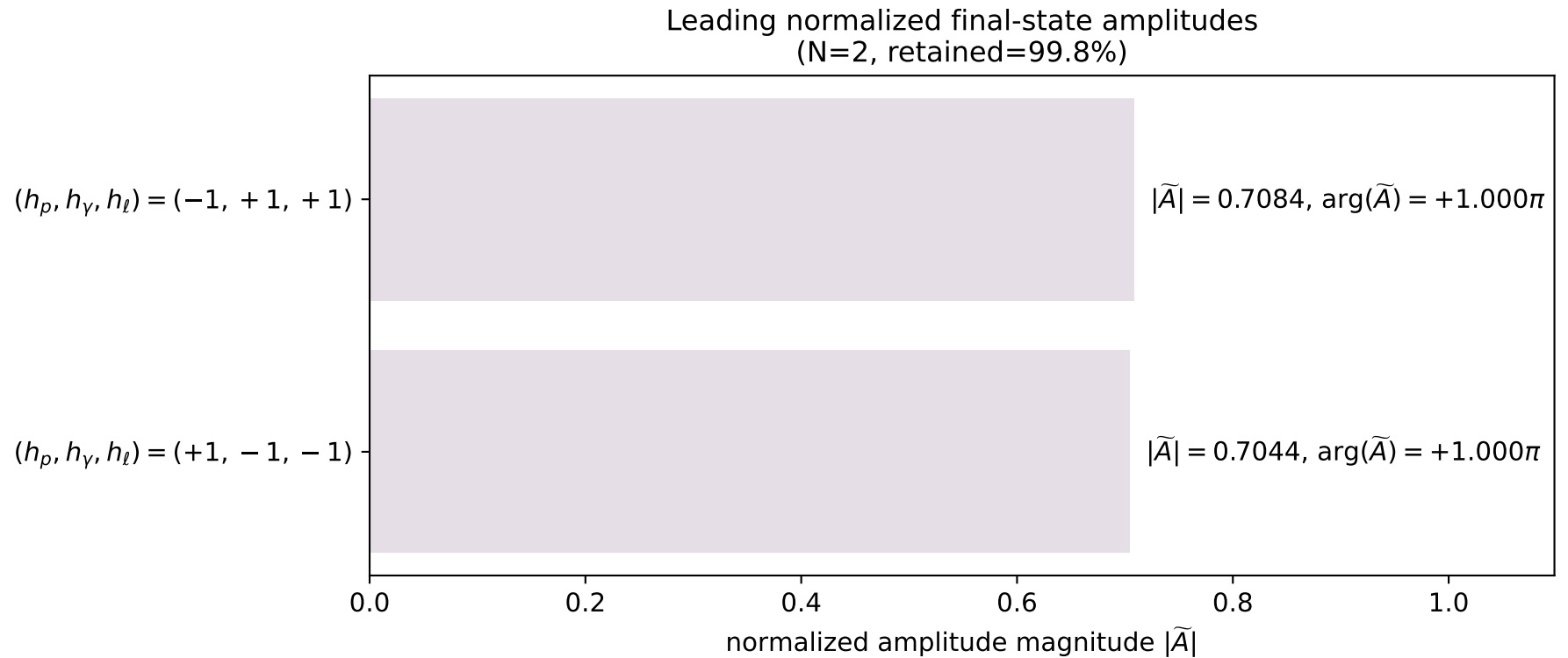


dghz_mixing_angles_polarization_03_local_minimum_267
 $d_{\{\rm GHZ\}}=0.000191782$
 region: polarization_cluster_03_local_minimum_267
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0267
 lepton mixing angle: $\alpha_e=2.647731$ rad
 incoming lepton: $-0.880509|+\rangle +0.47403|-\rangle$
 proton mixing angle: $\alpha_p=1.0728653$ rad
 incoming proton: $0.477609|+\rangle +0.878573|-\rangle$

$s=24.7744$, $\sqrt{s}=4.97739$, $|\vec{P}|=2.40031$, $|\vec{P}'|=1.37624$
 $\theta_{P'}=0.00145035$, $\theta_\gamma=3.13587$, $\phi_{P'}=3.13934$, $\phi_\gamma=0$
 $E_\gamma=2.34405$, $\theta_{l'}=0.0117929$, $\phi_{l'}=3.14199$
 $Q^2=9.29214$, $x_B=0.394536$, $t=-0.217746$, $W^2=15.1398$, $y=0.985667$

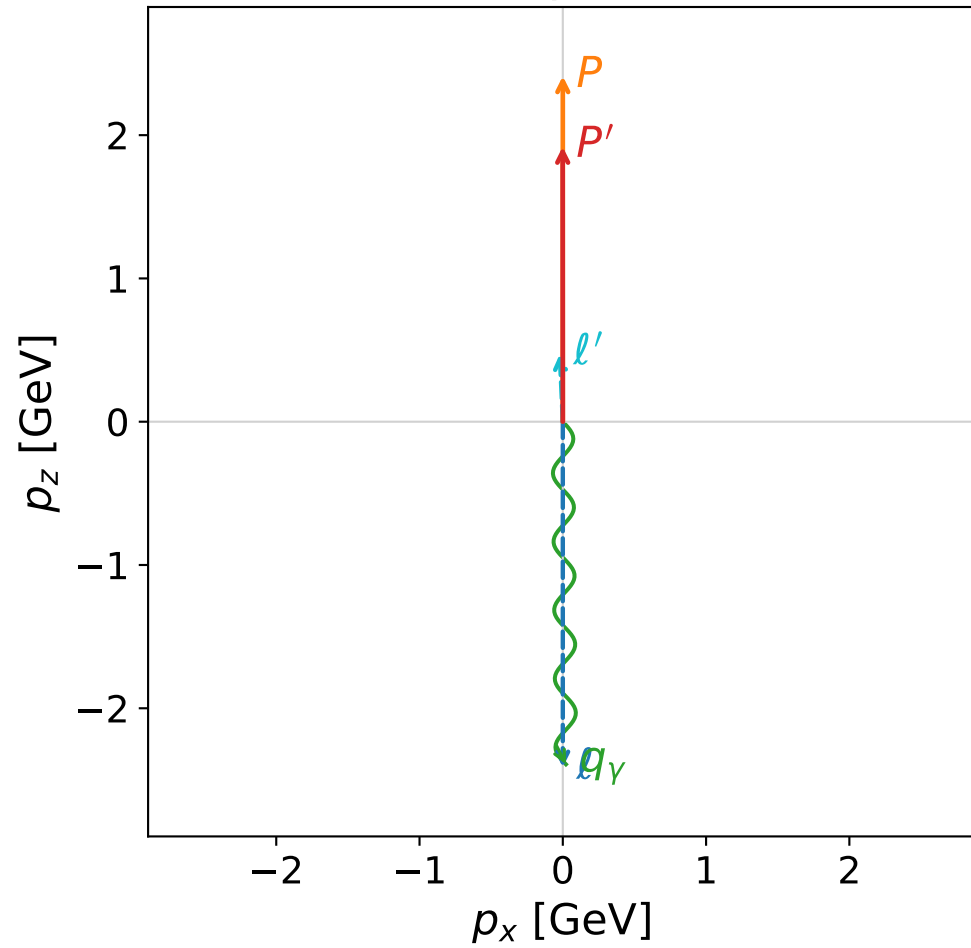
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4003$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4003$
 P $E= 2.5771$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4003$
 l' $E= 0.9678$ $p_x= -0.0114$ $p_y= -0.0000$ $p_z= 0.9678$
 P' $E= 1.6655$ $p_x= -0.0020$ $p_y= 0.0000$ $p_z= 1.3762$
 q_γ $E= 2.3441$ $p_x= 0.0134$ $p_y= 0.0000$ $p_z= -2.3440$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.245%, conserve 99.755%
 $R_{\text{flip/conserve}}=2.460\text{e-}03$; $I_h=-1.114\text{e-}03$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=2.880\text{e-}06$; $I_{TL}=-4.054\text{e-}04$

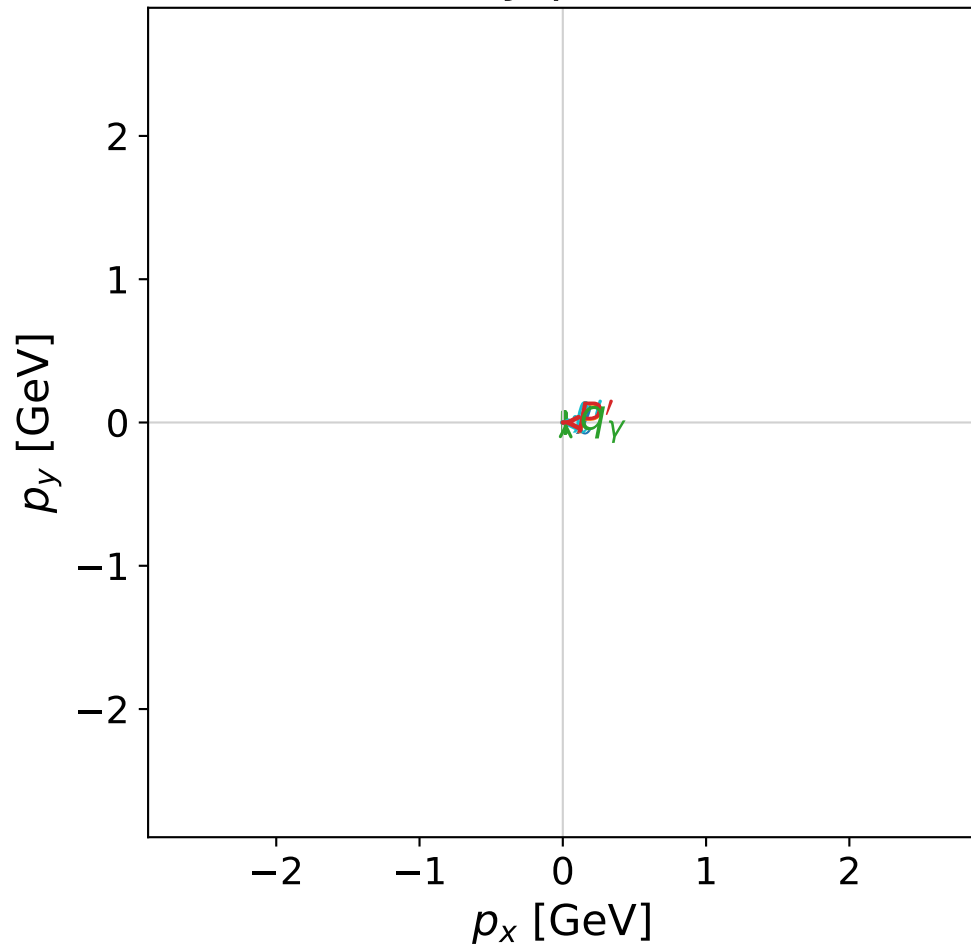


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



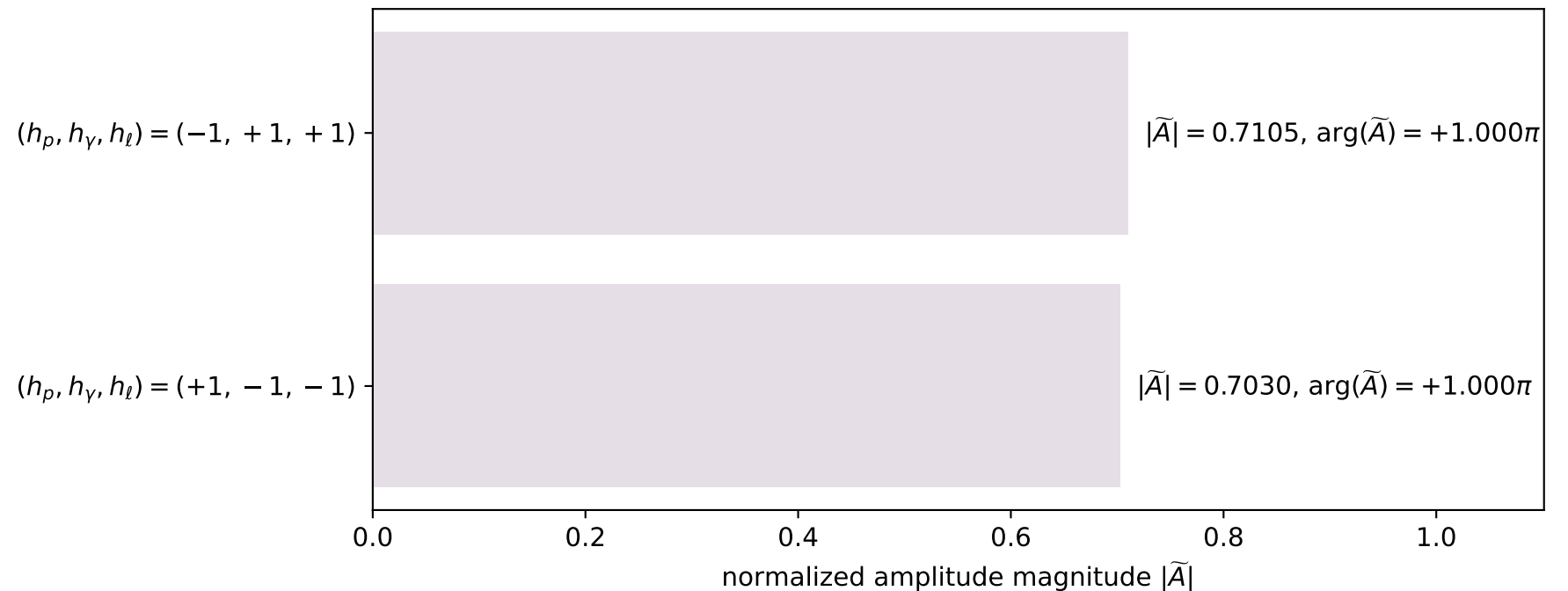
dghz_mixing_angles_polarization_03_local_minimum_316
 $d_{\text{GHZ}}=0.00040247$
 region: polarization_cluster_03_local_minimum_316
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0316
 lepton mixing angle: $\alpha_e=2.555944$ rad
 incoming lepton: $-0.833354|+\rangle + 0.55274|-\rangle$
 proton mixing angle: $\alpha_p=0.97424173$ rad
 incoming proton: $0.561796|+\rangle + 0.827276|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.91934$
 $\theta_{p'}=0.000445521$, $\theta_\gamma=3.13237$, $\phi_{p'}=3.14125$, $\phi_\gamma=6.28316$
 $E_\gamma=2.39134$, $\theta_{l'}=0.0448925$, $\phi_{l'}=3.14158$
 $Q^2=4.55525$, $x_B=0.190186$, $t=-0.0386979$, $W^2=20.2762$, $y=0.993012$

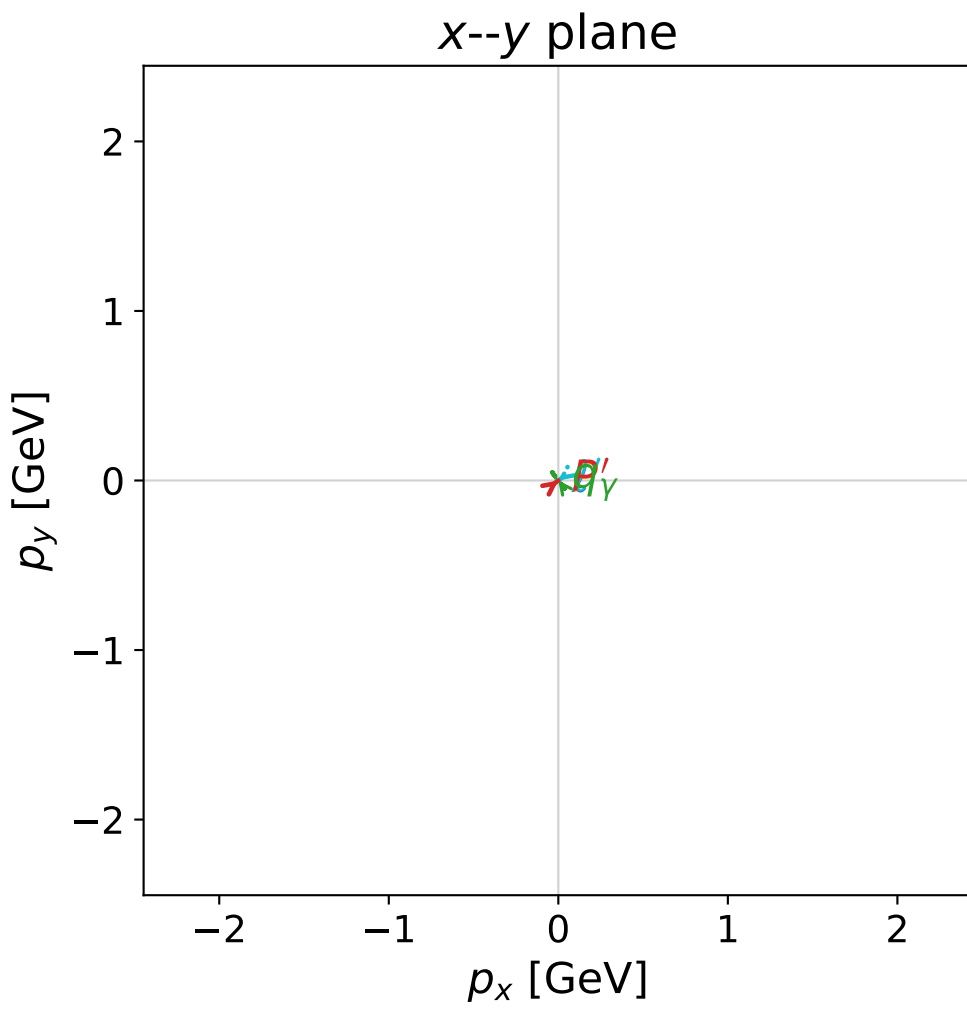
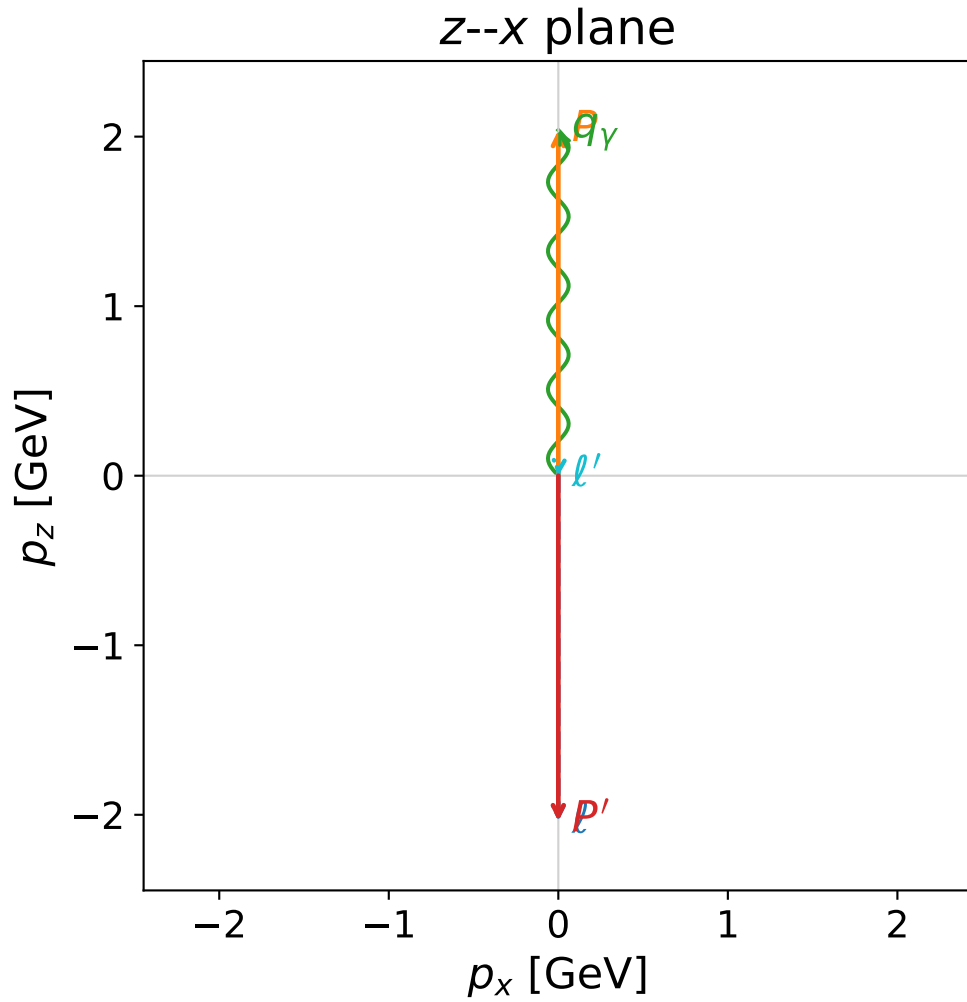
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.4724$ $p_x= -0.0212$ $p_y= 0.0000$ $p_z= 0.4719$
 P' $E= 2.1363$ $p_x= -0.0009$ $p_y= 0.0000$ $p_z= 1.9193$
 q_γ $E= 2.3913$ $p_x= 0.0221$ $p_y= -0.0000$ $p_z= -2.3912$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.070%, conserve 99.930%
 $R_{\text{flip/conserve}}=7.015\text{e-}04$; $I_h=-1.617\text{e-}04$
 virtual-photon powers: T 99.997%, L 0.003%
 $R_L/T=2.547\text{e-}05$; $I_{TL}=-4.788\text{e-}04$

Leading normalized final-state amplitudes
 (N=2, retained=99.9%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



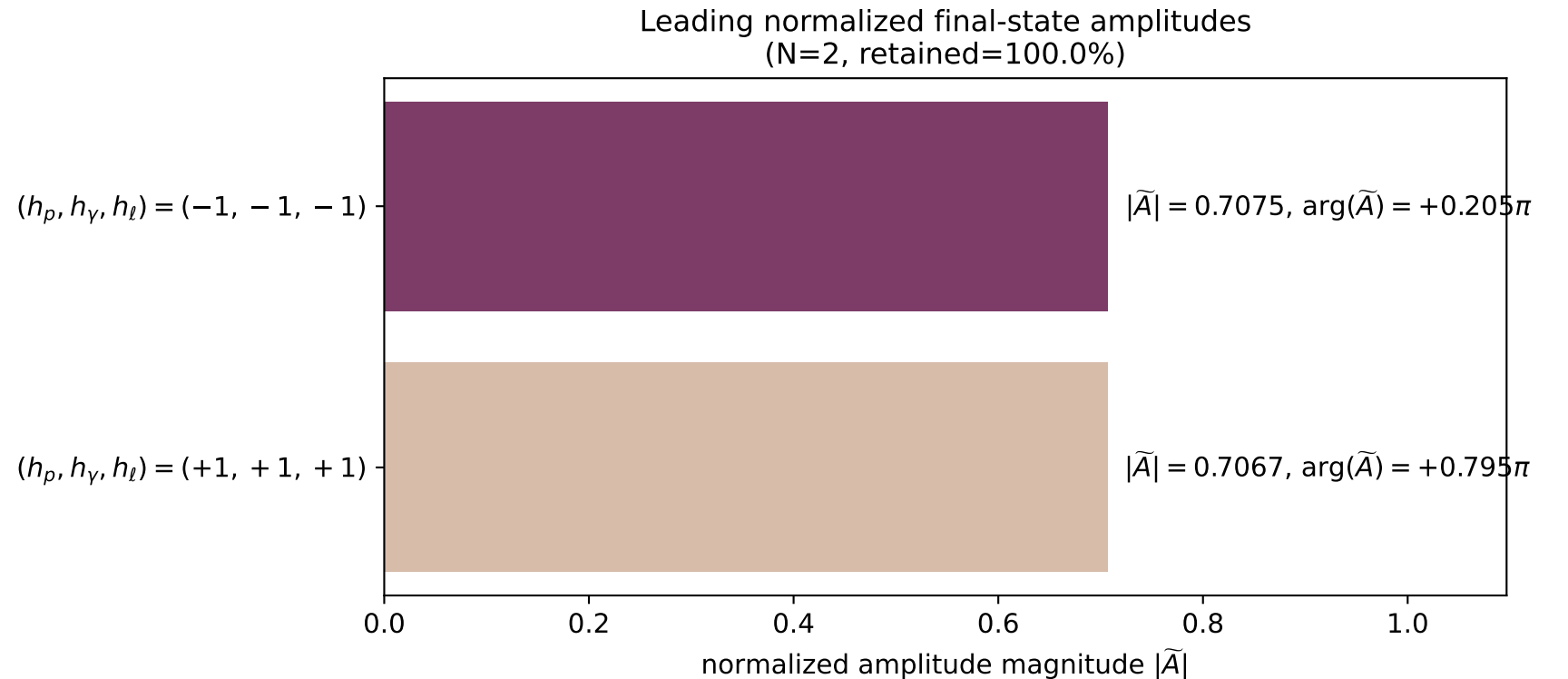
dghz_mixing_angles_polarization_03_local_minimum_357
 $d_{\{\text{rm GHZ}\}}=0.00125916$
 region: polarization_cluster_03_local_minimum_357
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0357
 lepton mixing angle: $\alpha_e=2.3565227$ rad
 incoming lepton: $-0.707339|+\rangle +0.706875|-\rangle$
 proton mixing angle: $\alpha_p=0.78632898$ rad
 incoming proton: $0.706448|+\rangle +0.707765|-\rangle$

$s=18.3383$, $\sqrt{s}=4.28232$, $|\vec{P}|=2.03843$, $|\vec{P}'|=2.03709$
 $\theta_p=3.14158$, $\theta_\gamma=2.79619\text{e-}05$, $\phi_p=0.644969$, $\phi_\gamma=0.64563$
 $E_\gamma=2.03832$, $\theta_{\ell'}=3.06993$, $\phi_{\ell'}=3.78699$
 $Q^2=0.000426632$, $x_B=2.44525\text{e-}05$, $t=-16.6098$, $W^2=18.3268$, $y=0.999369$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

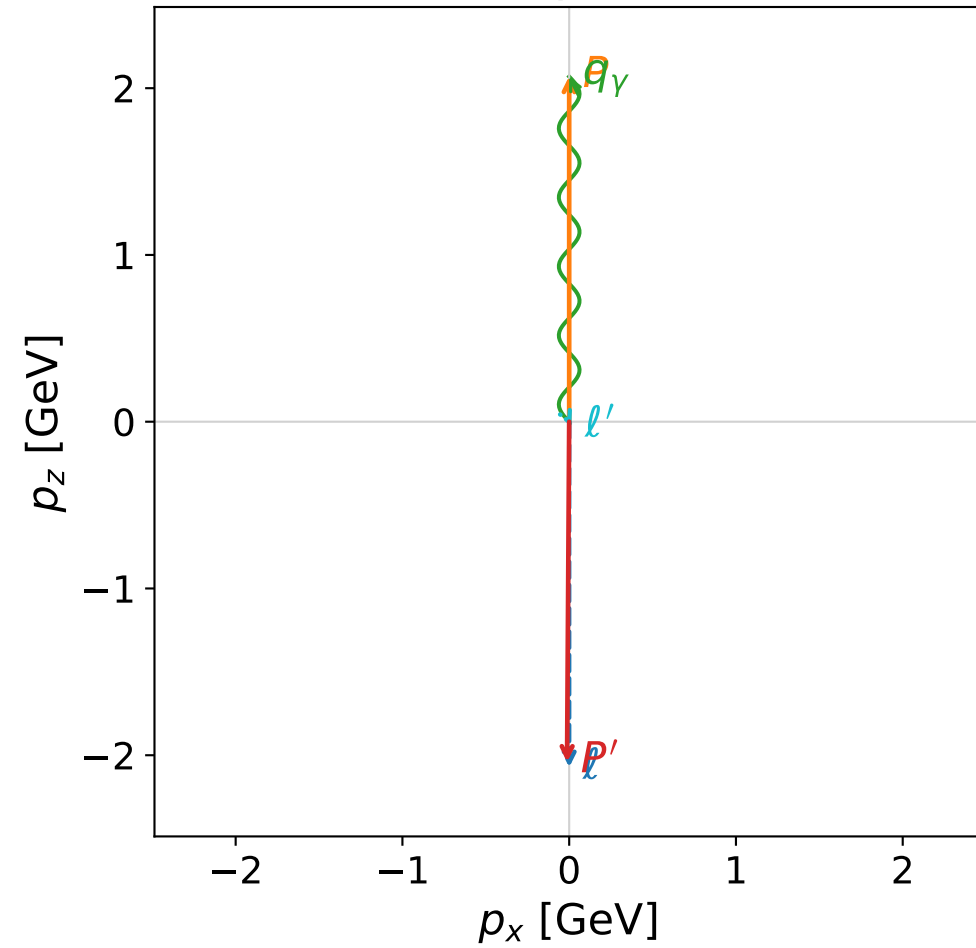
Particle	E	p_x	p_y	p_z
ℓ	2.0384	0.0000	0.0000	-2.0384
P	2.2439	0.0000	0.0000	2.0384
ℓ'	0.0013	-0.0001	-0.0001	-0.0012
P'	2.2427	0.0000	0.0000	-2.0371
q_γ	2.0383	0.0000	0.0000	2.0383

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.005%, conserve 99.995%
 $R_{\text{flip/conserve}}=5.071\text{e-}05$; $I_h=+2.567\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.059\text{e-}12$; $I_{TL}=-6.654\text{e-}11$

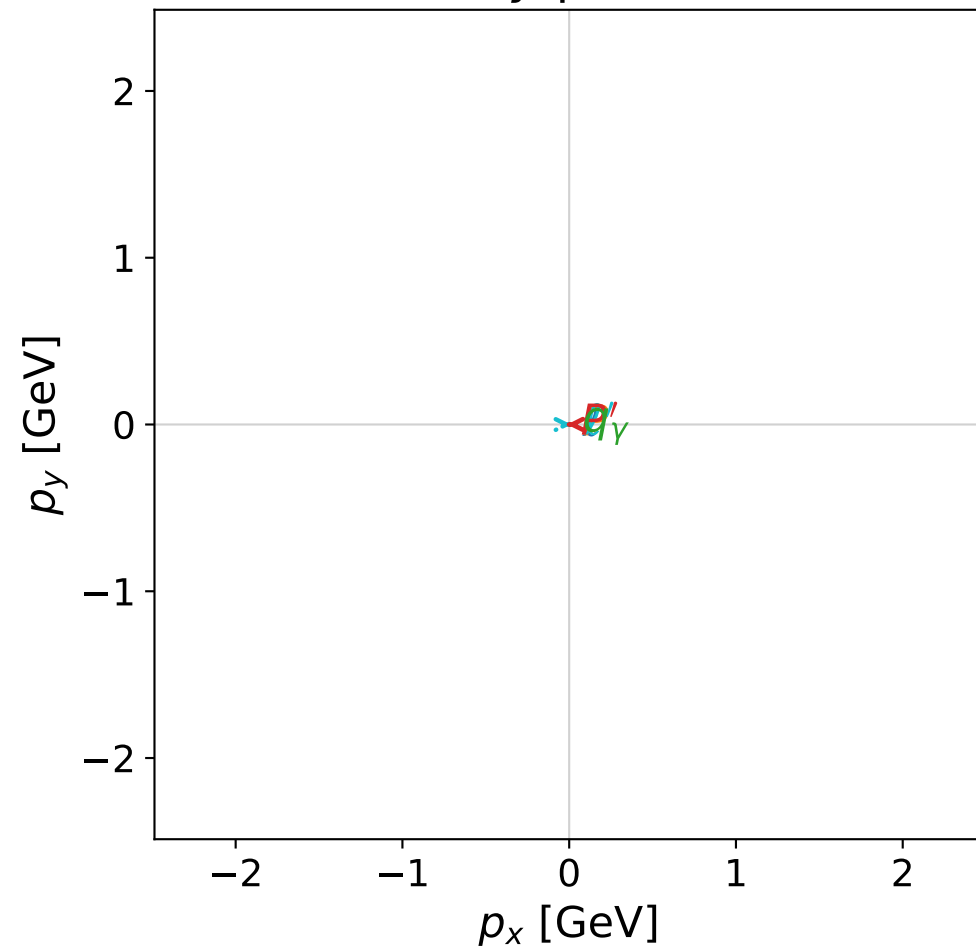


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

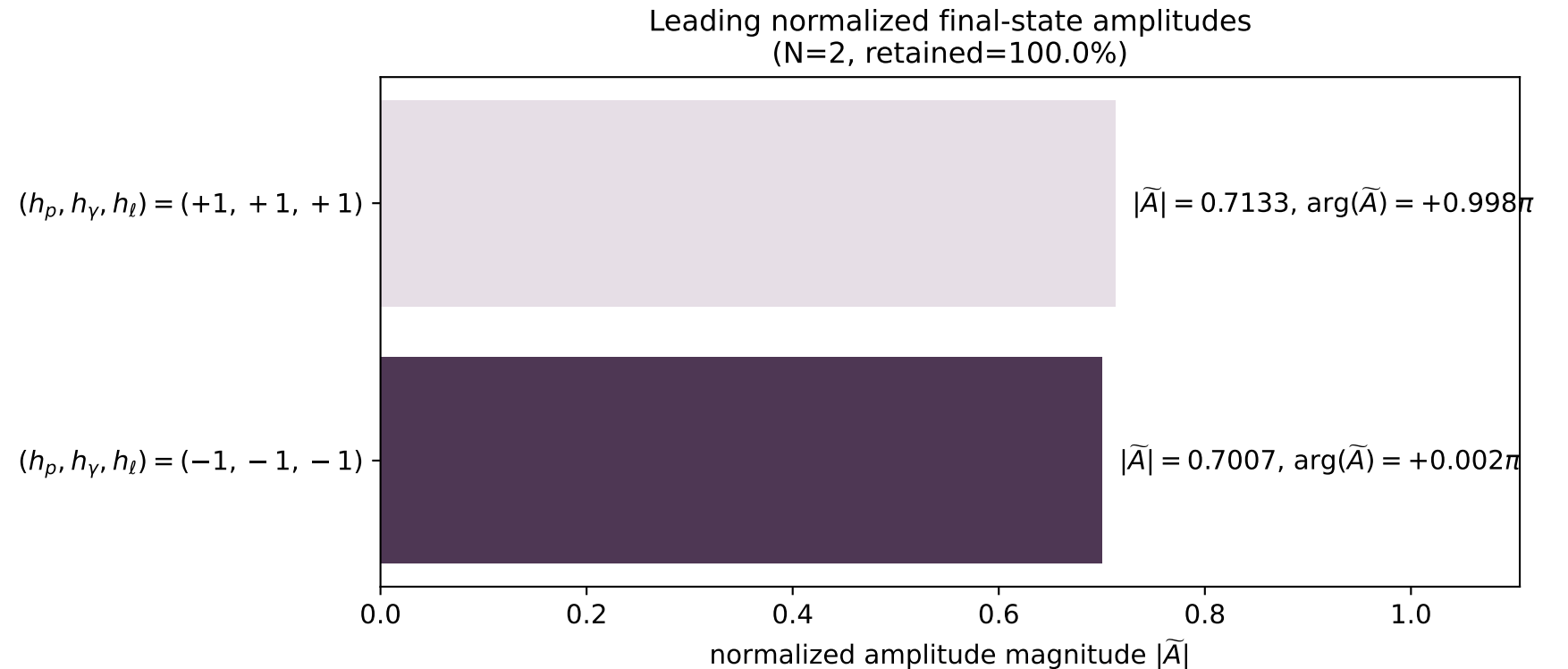


dghz_mixing_angles_polarization_03_local_minimum_372
 $d_{\text{GHZ}}=0.0286865$
 region: polarization_cluster_03_local_minimum_372
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0372
 lepton mixing angle: $\alpha_e=2.3560927$ rad
 incoming lepton: $-0.707035|+\rangle +0.707179|-\rangle$
 proton mixing angle: $\alpha_p=0.77718399$ rad
 incoming proton: $0.712891|+\rangle +0.701275|-\rangle$

$s=18.8971$, $\sqrt{s}=4.34708$, $|\vec{P}|=2.07234$, $|\vec{P}'|=2.04126$
 $\theta_{p'}=3.13533$, $\theta_\gamma=0$, $\phi_{p'}=3.14439$, $\phi_\gamma=1.40154$
 $E_\gamma=2.06954$, $\theta_{l'}=2.71784$, $\phi_{l'}=0.00279457$
 $Q^2=0.0114083$, $x_B=0.000642414$, $t=-16.9207$, $W^2=18.6269$, $y=0.985637$

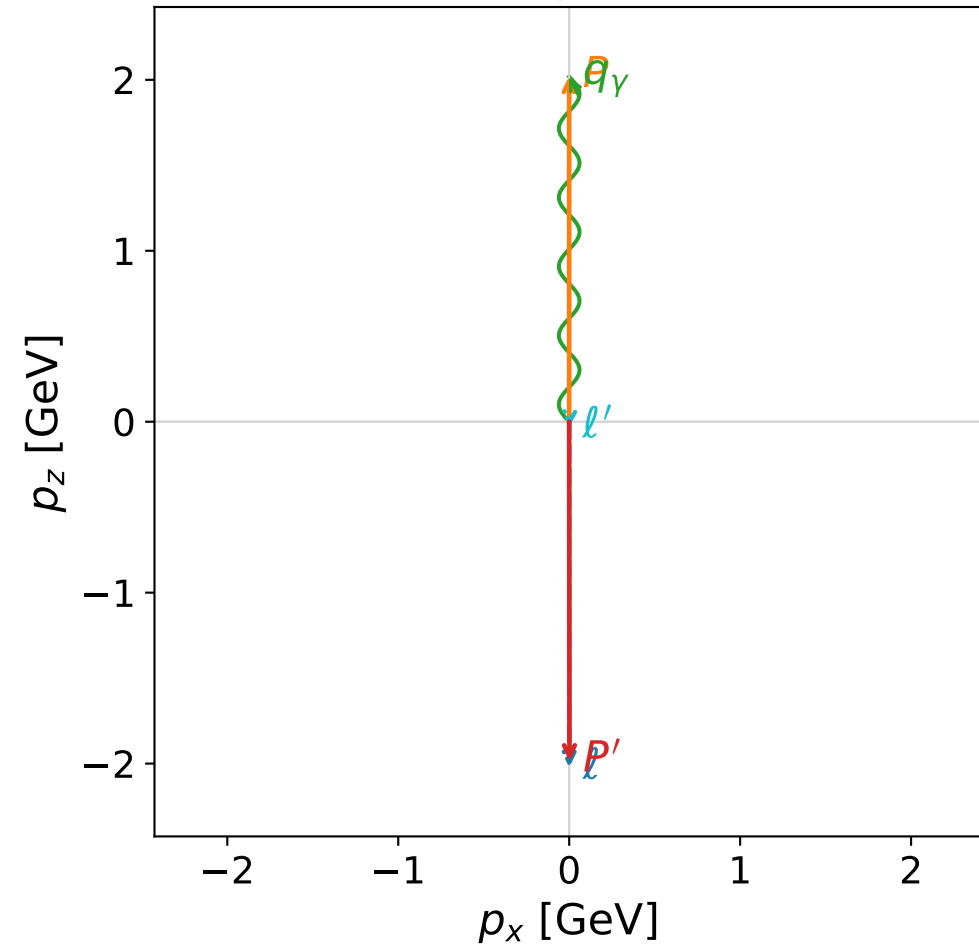
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0723$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0723$
 P $E= 2.2747$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0723$
 l' $E= 0.0311$ $p_x= 0.0128$ $p_y= 0.0000$ $p_z= -0.0283$
 P' $E= 2.2465$ $p_x= -0.0128$ $p_y= -0.0000$ $p_z= -2.0412$
 q_γ $E= 2.0695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0695$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.000%, conserve 100.000%
 $R_{\text{flip/conserve}}=3.037\text{e-}06$; $I_h=+2.445\text{e-}08$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.830\text{e-}08$; $I_{TL}=+9.046\text{e-}06$

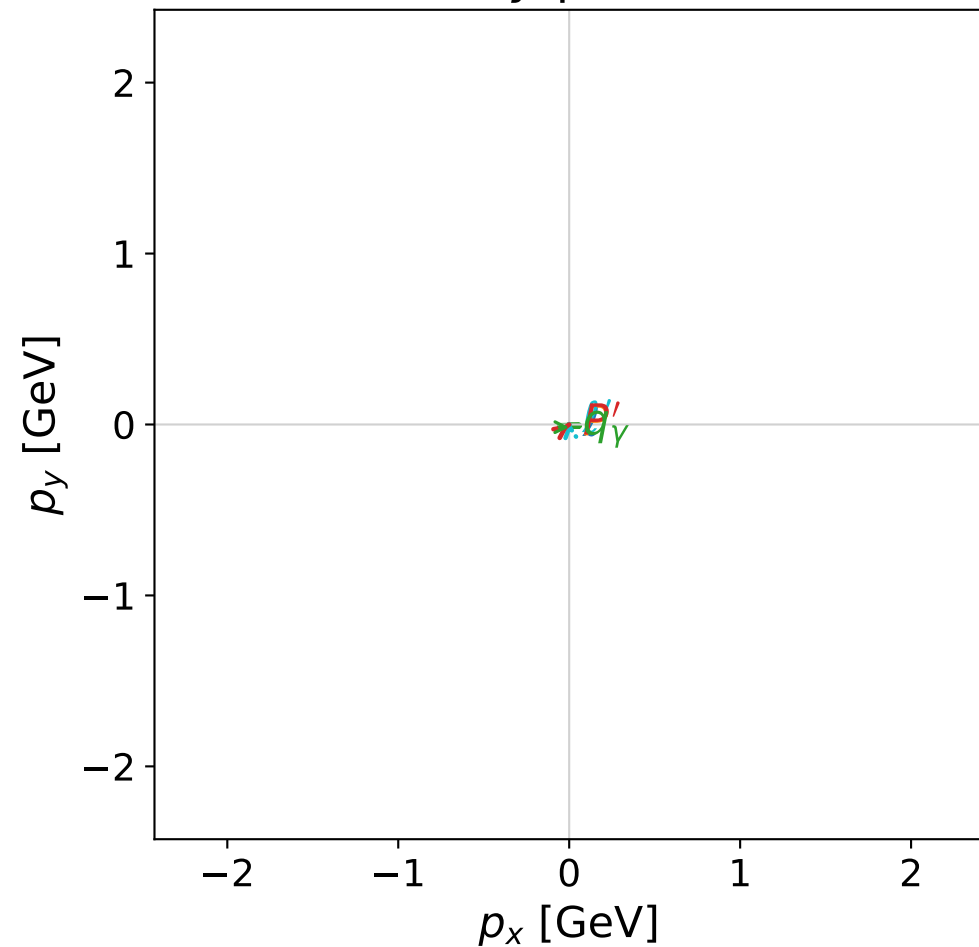


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_374
 $d_{\text{GHZ}}=0.0335652$
 region: polarization_cluster_03_local_minimum_374
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0374
 lepton mixing angle: $\alpha_e=2.356188$ rad
 incoming lepton: $-0.707102|+\rangle + 0.707111|-\rangle$
 proton mixing angle: $\alpha_p=0.79277071$ rad
 incoming proton: $0.701874|+\rangle + 0.712301|-\rangle$

$s=18.0658$, $\sqrt{s}=4.25039$, $|\vec{P}|=2.02169$, $|\vec{P}'|=1.98608$
 $\theta_{p'}=3.14052$, $\theta_\gamma=0.00808957$, $\phi_{p'}=0.622432$, $\phi_\gamma=4.72104$
 $E_\gamma=2.01834$, $\theta_{l'}=2.70049$, $\phi_{l'}=1.69454$
 $Q^2=0.0137918$, $x_B=0.000816221$, $t=-16.0612$, $W^2=17.7632$, $y=0.983193$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0217$
 P $E= 2.2287$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0217$
 l' $E= 0.0356$ $p_x= -0.0019$ $p_y= 0.0151$ $p_z= -0.0322$
 P' $E= 2.1964$ $p_x= 0.0017$ $p_y= 0.0012$ $p_z= -1.9861$
 q_γ $E= 2.0183$ $p_x= 0.0001$ $p_y= -0.0163$ $p_z= 2.0183$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.000%, conserve 100.000%
 $R_{\text{flip/conserve}}=2.406\text{e-}06$; $I_h=+1.478\text{e-}08$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=2.609\text{e-}08$; $I_{TL}=+2.400\text{e-}07$

